

Exploratory Analysis of Rainfall Data in India for Agriculture

9.2 - DISADVANTAGES:

1. **Depends on Data Quality:** The prediction accuracy depends on the correctness of historical weather data. Incorrect or incomplete data can give wrong results.
2. **Not 100% Accurate:** Weather is a natural phenomenon and can change suddenly. Therefore, the system cannot guarantee perfectly accurate rainfall prediction.
3. **Limited to Available Parameters:** The model only considers selected features like temperature, humidity, and pressure. Other environmental factors may also affect rainfall but are not included.
4. **Requires User Input:** The user must manually enter weather values. If wrong values are entered, the prediction will also be incorrect.
5. **Internet/Computer Requirement:** Farmers need a smartphone or computer to use the system, which may not be available in rural areas.
6. **Model Needs Regular Updates:** The machine learning model must be retrained periodically with new data to maintain accuracy.
7. **Local Weather Variation:** The system may not perfectly predict rainfall for very small regions because weather varies from place to place.
8. **Cannot Handle Extreme Climate Events:** Sudden cyclones, storms, or unexpected climate changes may not be predicted properly.